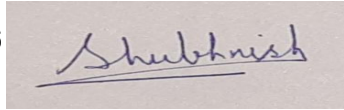


Department of Computing and Mathematics
Computing and Digital Technology Postgraduate
Programmes Terms of Reference CoverSheet

Student name:	Shubhnish Kumar
University id:	25933686
Academic Supervisor	Connah Kendrick
Project Title	Scalable AI-Driven NPC Simulation System for Unreal Engine 5
Degree Title:	MSc Computer Games Development
Project unit code;	6G7V0007
Credit rating:	60
Start date:	01/06/26
ToR date:	29/05/26
Intended submission date:	11/09/26

Signature and date student:	29/05/26 

Aim

To develop a scalable NPC simulation system in Unreal Engine 5 that enables hundreds of interactable non-player characters (NPCs) with dynamic behaviors, persistent states, procedural schedules, and optimized real-time AI to create a believable living game world.

Objectives

1. Design and implement a large-scale NPC management framework using Unreal Engine 5.
 2. Develop dynamic NPC behavior systems using StateTree, AI logic, and procedural decision-making.
 3. Implement optimized crowd simulation techniques using Unreal Engine 5 Mass Entity framework.
 4. Create interactive NPC systems including dialogue, schedules, memory, and player interaction.
 5. Develop spawn/despawn and abstract simulation systems to improve performance for large populations.
 6. Optimize animation, navigation, and AI processing to support hundreds of NPCs simultaneously.
 7. Evaluate system performance, scalability, and realism within an open-world test environment.
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Background

Modern open-world games require realistic and immersive NPC populations that can interact dynamically with players and the environment. Traditional AI systems become computationally expensive when handling large numbers of active NPCs simultaneously. This project aims to address this challenge by combining scalable AI architectures, procedural simulation, and Unreal Engine 5 technologies such as Mass Entity and StateTree to create an efficient and believable living-world system.

Scope of the Project

The project will include:

- Development in Unreal Engine 5
- AI-driven NPC behavior systems
- Crowd simulation and optimization
- NPC interaction and dialogue systems
- Persistent NPC state management
- Performance profiling and scalability testing

The project will not include:

- Multiplayer networking
 - Full voice AI integration
 - Photorealistic cinematic systems
 - Advanced machine learning training pipelines
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Expected Deliverables

- A fully functional Unreal Engine 5 prototype
 - Scalable NPC simulation framework
 - Interactive NPC system with behaviors and schedules
 - Performance evaluation report
 - Technical documentation and implementation details
 - Final dissertation/project report
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Technologies and Tools

- Unreal Engine 5

- C++
 - Blueprint System
 - Mass Entity Framework
 - StateTree AI
 - Behavior Trees
 - Git/GitHub
 - Visual Studio
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Research Areas

- Artificial Intelligence in Games
 - Crowd Simulation
 - Procedural Systems
 - Real-Time Optimization
 - Open-World Game Design
 - Entity Component Systems (ECS)
-

Proposed Methodology

1. Research existing large-scale NPC systems in modern games.
 2. Design NPC architecture and simulation framework.
 3. Implement core AI and interaction systems.
 4. Integrate optimization techniques for large populations.
 5. Conduct testing and performance benchmarking.
 6. Evaluate realism, scalability, and system efficiency.
 7. Produce documentation and final evaluation.
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Success Criteria

The project will be considered successful if it can:

- Support hundreds of NPCs simultaneously
- Maintain stable real-time performance
- Provide meaningful NPC interactions
- Demonstrate scalable AI architecture
- Create a believable living-world simulation

